

INSTACART RETAIL SALES & CUSTOMER INSIGHTS

1. Executive Summary

This project analyzes Instacart retail order data to understand customer purchasing behavior, reorder patterns, product demand, department performance, and order timing.

The analysis covers **3.42 million+ orders, 206K+ customers, and 49K+ products**. Python was used for data preparation and analysis, Excel was used to create MIS-style reports, and Power BI was used to develop an interactive dashboard for KPI monitoring and business insights.

The analysis identified **Produce as the highest-volume department with 9.48M purchases**, while **High-Frequency customers recorded the highest reorder rate of 69.57%**. Medium-Frequency customers represented the largest customer segment at **46.43%**, and **10 AM recorded the highest order activity at 8.43%**.

2. Business Problem

Retail teams need to understand customer purchasing and reorder behavior to identify high-value customer segments, high-demand departments, and peak ordering periods for effective performance monitoring and planning.

The project focuses on converting large-scale transaction data into structured MIS reports and actionable retail insights.

3. Project Objectives

The key objectives of this project were to:

- Analyze overall order and customer activity.
- Understand customer purchase frequency and reorder behavior.
- Identify high-volume departments and products.
- Compare reorder rates across departments and customer segments.
- Identify peak ordering days and hours.
- Develop structured Excel MIS reports.
- Build an interactive Power BI dashboard for KPI and performance monitoring.

4. Dataset Overview

The project uses the Instacart Online Grocery Shopping dataset containing multiple relational datasets covering orders, products, departments, aisles, and order-product relationships.

Dataset Components

Dataset	Purpose
Orders	Customer order history and order timing
Products	Product-level information
Departments	Department classification
Aisles	Product aisle classification
Prior Orders	Historical product-level order activity
Train Orders	Product-level order activity for the training set

Dataset Scale

- **Total Orders:** 3,421,083
- **Total Customers:** 206,209
- **Total Products:** 49,688
- **Product-level Order Records:** 32M+ across prior and train data

5. Tools & Technologies

Python

- Pandas
- NumPy
- Data cleaning
- Data transformation
- Exploratory Data Analysis
- Aggregation and statistical calculations

Microsoft Excel

- MIS reporting
- Pivot-style analysis
- KPI summaries
- Data visualization
- Department and product analysis

Power BI

- Interactive dashboard development
- KPI cards
- Customer segmentation
- Department analysis
- Reorder analysis
- Order timing analysis
- Business insights

6. Data Preparation & Quality Checks

The datasets were reviewed and prepared before analysis.

Key data-quality checks included:

- Identifying missing values.
- Checking duplicate records.
- Reviewing column data types.
- Validating categorical fields.
- Creating calculated metrics.
- Combining relevant datasets for product and department analysis.
- Preparing summarized datasets for Excel and Power BI.

Data Quality Results

- **Orders:** Missing values were present only in days_since_prior_order, corresponding to first-order records.
- **Products:** No missing values identified.
- **Departments:** No missing values identified.
- **Aisles:** No missing values identified.
- Duplicate checks for the main reference datasets returned **zero duplicates**.

7. Key Performance Indicators

KPI	Result
Total Orders	3,421,083
Total Customers	206,209
Total Products	49,688
Average Products per Order	10.09
Overall Reorder Rate	58.97%

These KPIs provide a high-level view of the scale of customer activity and repeat purchasing behavior.

8. Customer Segmentation

Customers were segmented according to purchase frequency into three groups:

Customer Segment	Customers	Share
Low Frequency	59,741	28.97%
Medium Frequency	95,737	46.43%
High Frequency	50,731	24.60%

Reorder Rate by Segment

Segment	Reorder Rate
Low Frequency	27.74%
Medium Frequency	48.57%
High Frequency	69.57%

Key Finding

High-Frequency customers recorded the highest reorder rate at 69.57%, while Medium-Frequency customers formed the largest segment at 46.43%.

This highlights the importance of monitoring both high-frequency customers for retention and the larger medium-frequency segment for potential engagement opportunities.

9. Department Purchase Analysis

The analysis identified the departments generating the highest purchase volumes.

Top Departments by Purchase Volume

Department	Purchases
Produce	9,479,291
Dairy Eggs	5,414,016
Snacks	2,887,550
Beverages	2,690,129
Frozen	2,236,432
Pantry	1,875,577
Bakery	1,176,787
Canned Goods	1,068,058
Deli	1,051,249
Dry Goods Pasta	866,627

Key Finding

Produce recorded the highest purchase volume with 9.48M purchases, making it the largest department by transaction volume in the analyzed dataset.

10. Department Reorder Analysis

Reorder rates were calculated across all 21 departments.

Highest Reorder Rates

Department	Reorder Rate
Dairy Eggs	67.00%
Beverages	65.35%
Produce	64.99%
Bakery	62.81%
Deli	60.77%

Key Finding

Dairy Eggs recorded the highest department-level reorder rate at 67.00%.

This indicates strong repeat-purchase behavior within the department.

11. Product Analysis

Product-level purchasing behavior was analyzed to identify highly purchased and highly reordered products.

Top Purchased Products

Product	Purchases
Banana	472,565
Bag of Organic Bananas	379,450
Organic Strawberries	264,683
Organic Baby Spinach	241,921
Organic Hass Avocado	213,584

Key Finding

Banana was the most purchased product with 472,565 purchases.

The product analysis also identified highly reordered products, with several milk and dairy products recording reorder rates above 85%.

12. Order Timing Analysis

Order activity was analyzed by day of week and hour of day.

Order Activity by Day

Day	Share of Orders
Day 0	17.56%
Day 1	17.17%
Day 2	13.66%
Day 3	12.77%
Day 4	12.46%
Day 5	13.25%
Day 6	13.12%

Key Finding

Day 0 recorded the highest share of orders at 17.56%.

Order Activity by Hour

The analysis showed that ordering activity was concentrated during daytime hours.

10 AM recorded the highest order activity with 271,885 orders, representing 8.43% of total orders.

This provides a useful reference point for understanding peak customer activity.

13. MIS Reporting

An Excel MIS workbook was created to organize the analysis into business-focused reporting sections.

MIS Workbook Sections

1. KPI Summary
2. Customer Segments
3. Department Analysis
4. Product Analysis
5. Order Timing
6. Reorder Analysis

The reports provide summarized metrics that can be used to monitor customer behavior, department performance, product trends, and order activity.

14. Power BI Dashboard

An interactive Power BI dashboard was developed to provide a consolidated view of the retail analysis.

Dashboard Components

- Total Orders
- Total Customers
- Total Products
- Average Products per Order
- Overall Reorder Rate
- Customer Distribution by Frequency
- Top Departments by Purchase Volume
- Reorder Rate by Department
- Order Activity by Hour
- Order Distribution by Day
- Top Products by Reorder Rate
- Key Business Insights

The dashboard enables users to explore the major performance indicators and behavioral patterns through interactive visualizations.

15. Key Business Findings

The analysis produced the following major findings:

1. **3.42M+ orders** were analyzed across **206K+ customers**.
2. **Produce** generated the highest purchase volume with **9.48M purchases**.
3. **Dairy Eggs** had the highest department reorder rate at **67.00%**.
4. **High-Frequency customers** recorded the highest reorder rate at **69.57%**.
5. **Medium-Frequency customers** represented the largest customer segment at **46.43%**.
6. **Banana** was the most purchased product with **472,565 purchases**.
7. **10 AM** recorded the highest order activity with **271,885 orders / 8.43%**.
8. The overall reorder rate was **58.97%**.
9. The average order contained **10.09 products**.

16. Business Recommendations

Based on the analysis:

- Monitor **high-reorder departments** such as Dairy Eggs and Beverages to understand repeat-purchase behavior.
- Focus customer engagement efforts on the **Medium-Frequency segment**, which represents the largest customer group.
- Continue monitoring **High-Frequency customers** because of their stronger reorder behavior.
- Use peak ordering periods such as **10 AM** as a reference for operational planning.
- Monitor high-volume departments such as **Produce** to support inventory and demand planning.
- Track highly reordered products to understand recurring customer needs.

17. Project Outcome

The project transformed large-scale Instacart transaction data into structured analytical outputs using **Python, Excel, and Power BI**.

The final deliverables included:

- Cleaned and analyzed datasets.
- Excel-based MIS reports.
- Customer segmentation analysis.
- Department and product performance analysis.
- Reorder behavior analysis.
- Order timing analysis.
- Interactive Power BI dashboard.

The project demonstrates the ability to move from

raw transactional data → data preparation → analysis → MIS reporting → dashboard development → business insights.

18. Skills Demonstrated

- Data Cleaning & Validation
- Exploratory Data Analysis
- Customer Segmentation
- Retail Analytics
- Reorder Analysis
- KPI Development
- MIS Reporting
- Data Visualization
- Business Insight Generation
- Python/Pandas
- Excel
- Power BI
- Dashboard Development

19. Project Deliverables

GitHub

Repository: <https://github.com/Lavanya-DA/instacart-data-analysis>

Files Included:

- Python analysis notebook
- Excel MIS workbook
- Power BI dashboard
- Project documentation / README

20. Conclusion

The Instacart Retail Sales & Customer Insights project demonstrates a complete data analytics workflow using large-scale retail transaction data. The analysis identified important patterns in customer frequency, reorder behavior, department performance, product demand, and order timing.

By combining Python-based analysis, Excel MIS reporting, and Power BI visualization, the project converts complex transactional data into structured metrics and business-focused insights that can support retail performance monitoring and decision-making.